Building Swift ROS2 Drivers for Other Distros

Last edited by **Joshua Marshall (JMARSHAL)** Jan 22, 2026, 3:53 p.m. EST

The current official swiftnav-ros2 driver for the Duro GPS device is written for ROS2 Humble distro. However, if we would like it to build it (from source) for a different distro, e.g., Jazzy, all we need are some minor modifications. Below are the steps to follow.

1. Open terminal.

2. Install common building tools (if not yet installed):

```
sudo apt update
sudo apt install -y \
  git cmake build-essential \
  python3-rosdep python3-colcon-common-extensions \
  python3-vcstool
sudo rosdep init 2>/dev/null || true
rosdep update
```

3. Install and build the dependency: `libsbp` (Swift's SBP library).

```
cd ~
git clone https://github.com/swift-nav/libsbp.git
cd libsbp
git checkout v4.11.0
cd c
git submodule update --init --recursive
mkdir -p build
cd build
cmake -DCMAKE_CXX_STANDARD=17 -DCMAKE_CXX_STANDARD_REQUIRED=ON -DCMAKE_CXX_EXTENSIONS=OFF ../
make
sudo make install
sudo ldconfig
```

4. clone the `swiftnav-ros2` repository:

```
cd /PATH/TO/YOUR/WORKSPACE/src
git clone https://github.com/swift-nav/swiftnav-ros2.git
```

5. Install remaining deps:

```
cd /PATH/TO/YOUR/WORKSPACE/
source /opt/ros/jazzy/setup.bash # if not yet written in .bashrc

sudo apt update
sudo apt install -y libserialport-dev

rosdep install --from-paths src --ignore-src -r -y
```

6. **IMPORTANT**: Fix C++ API syntax errors specifically for Jazzy

Edit: `src/swiftnav-ros2/src/logging/sbp_file_logger.cpp`

Add this include near the top:

```
#include <array>
```

Save, then edit `src/swiftnav-ros2/src/logging/sbp_file_logger.cpp`:

Find line 37, change

```
node->declare_parameter("enabled_publishers");
```

to

```
node->declare_parameter<std::vector<std::string>>(
  "enabled_publishers", std::vector<std::string>{});
```

and save.

7. Go back to workspace, and build:

```
colcon build --symlink-install
```

Comments